STUDY MATERIAL TO COMPLETE THE LEC. 3.6

SUMMARY MATERIAL FROM THE FINAL PART OF CHAPTER 12 OF THE BOOK BIOMEDICAL CELLULAR BIOLOGY: The cytoskeleton

Authors: Alfonso Calvo, José J. Abad, Francisco Expósito, Luis Montuenga

****

Learning objectives:

- **Know the structure and function of intermediate filaments.**

****

INTERMEDIATE FILAMENTS

Structure and assembly

Intermediate filaments have a diameter of 8-12 nm (an intermediate thickness between actin microfilaments and microtubules) and constitute polymers of great structural **stability** that participate in the maintenance of the cytoskeleton and in resistance to mechanical stress. Intermediate filaments have a series of morphofunctional characteristics that differentiate them from microtubules and microfilaments:

• Unlike microtubules (made up of tubulin) and microfilaments (which are actin polymers), intermediate filaments are made up of diverse proteins that give rise to different groups of filaments.

• The proteins that make up the intermediate filaments are filamentous and not globular (figure 1).

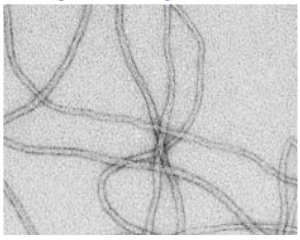


Figure 1. Electron microscopy image of intermediate filaments.

• Does not present polarity; That is, they do not have (+) or (-) ends, nor do they have a polymerization-depolymerization dynamic like that of microtubules or microfilaments.

• They do not require ATP or GTP for their polymerization.

• They do not have a motor function, but mainly structural.

More than 50 different proteins have been identified that can be part of intermediate filaments, which have been grouped into various classes, depending on their amino acid sequence and protein structure. Some intermediate filaments are specific to epithelial cells (and are called cytokeratins), while others are expressed in cells of mesenchymal origin or in nerve cells.

Although there are differences in molecular weight and amino acid sequence, the different intermediate filament proteins have a similar structural organization. The monomers (basic units) of these filaments are filamentous proteins. These monomeric units associate with each other in successive degrees of complexity until they form the filament (figure 2). The assembly and disassembly of intermediate filaments is regulated by phosphorylation; For example, the addition of phosphate to the filaments of the **nuclear laminae** of the nuclear envelope by the Cdk1-cyclin B complex causes the disassembly of these filaments.

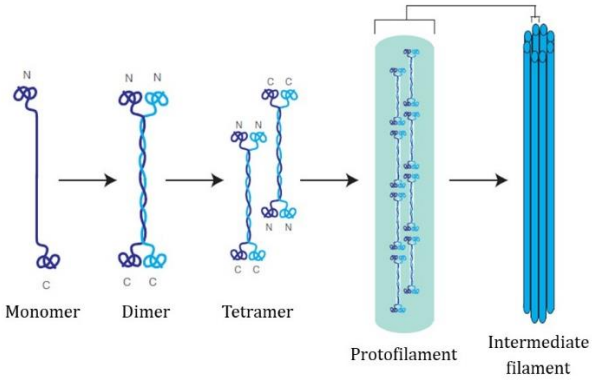


Figure 2. Scheme of the molecular structure of the intermediate filaments.

Intermediate filaments form a broad network in the cytoplasm that extends from the perinuclear zone (where they are most abundant) to the periphery (figure 3). These filaments help fix the cell nucleus in the characteristic position in which it is found in each type of cell. Intermediate filament networks are interconnected with microtubules, microfilaments, and proteins of the plasma membrane (such as integrins) and the nuclear membrane. A type of intermediate filament, which forms the **keratin** of epithelial cells (**tonofilaments**) is anchored to desmosomes and hemidesmosomes (figure 4).

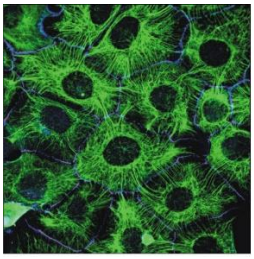


Figure 3. Immunofluorescence to detect intermediate filaments.

Intermediate filaments are essential for cells to maintain their structure and resist the mechanical stress to which they are subjected, especially intense for epithelial cells, and in particular that

of the skin. Mutations in cytokeratins cause different diseases, many of which are manifested in skin alterations.



Figure 4. Electron microscopy image of intermediate filaments associated to desmosomes.

Role of intermediate filaments in differential diagnosis

The distribution of certain types of intermediate filaments is specific to each cell type and constitutes an identity fingerprint of each tissue that can be used as a marker for differential diagnosis in histopathology (figure 5). For example, in the diagnosis of metastasis in patients in whom a primary tumor has not been identified, the localization of certain cytokeratins by immunohistochemistry (usually together with other markers) can allow us to determine the nature of the primary tumor, which may be essential to chhose a correct treatment.

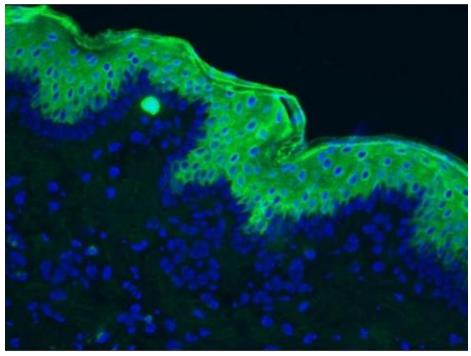


Figure 5. Immunofluorescence to detect keratin 10 in the epidermis.